CENG381 Stochastic Processes

Wald's Equality — Practice 1 (Answer Key)

Q1 Solution

a) Stopping time definition and examples:

A stopping time J is a random variable such that the event {J = n} depends only on the observed values X_1, ..., X_n — you can decide to stop at time n using only the information available up to time n (no 'peeking into the future').

Valid stopping time: J = first n such that S_n > 10. The decision depends only on X_1, ..., X_n.

NOT a valid stopping time: J = last n before the sum exceeds 10. Deciding which is 'last' requires knowledge of future X_{n+1}, X_{n+2}, ...

b) Dice rolls, stop at S_n >= 21:

```
X_bar = (1+2+3+4+5+6)/6 = 3.5
E[J] ~= 6  (given)

Wald's equality:  E[S_J] = X_bar * E[J] = 3.5 * 6 = 21
```

Intuitive: the game aims to reach 21, so the expected sum at stopping should be around 21. Wald's equality confirms this exactly (any overshoot is accounted for in E[J]~=6).

c) Exponential inter-arrivals, threshold c:

```
E[N(c)] ~= lambda*c  by the elementary renewal theorem
E[J] = E[N(c)+1] ~= lambda*c + 1

Wald's equality:  E[S_J] = (1/lambda) * (lambda*c + 1) = c + 1/lambda

Overshoot = E[S_J] - c = 1/lambda = E[X]
```

On average the sum overshoots the threshold c by exactly one mean inter-arrival time. This is the 'excess life' of the renewal process and is another way to see the inspection paradox.

Q2 Solution

a) Recurrence for theta:

Condition on the first flip:

```
theta = P(J < inf)
```

```
     = P(S_1=1)*1   +  P(S_1=-1)*P(two comebacks both succeed)
     = p * 1        +  q * theta^2
     = p + q*theta^2
```

The factor θ^2 appears because from S_1=−1 the gambler must first return to 0 (probability θ), then advance from 0 to 1 (another independent probability θ), giving θ^2.

b) Solving the quadratic:

```
q*theta^2 - theta + p = 0
(1-p)*theta^2 - theta + p = 0

Discriminant: 1 - 4*(1-p)*p = 1 - 4p + 4p^2 = (1-2p)^2
Roots: theta = [1 +- (1-2p)] / (2*(1-p))
  theta_1 = [1 + (1-2p)] / (2*(1-p)) = 2*(1-p)/(2*(1-p)) = 1
  theta_2 = [1 - (1-2p)] / (2*(1-p)) = 2p/(2*(1-p)) = p/q

For p > 1/2: p/q > 1, not a valid probability  =>  theta = 1
For p < 1/2: p/q < 1, theta = p/q < 1 (game ends with prob < 1)
```

c) E[J] and verification:

```
E[J] = p*1 + q*(1 + 2*E[J])
E[J] = p + q + 2*q*E[J]
E[J] - 2*q*E[J] = 1
E[J]*(1 - 2*q) = 1
E[J]*(2p - 1) = 1
E[J] = 1/(2p-1)

Wald's equality: E[X] = p*1 + q*(-1) = 2p-1
E[S_J] = (2p-1) * 1/(2p-1) = 1 = S_J  ✓

For p = 2/3:  q = 1/3,  theta = 1
  E[J] = 1/(2*(2/3)-1) = 1/(4/3-1) = 1/(1/3) = 3
  E[S_J] = 1  ✓

As p --> 1/2+:  E[J] = 1/(2p-1) --> +inf
```

```
(The game takes longer and longer as the bias vanishes.)
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Q3 Solution

a) Stopping rule A — Negative Binomial:

```
q = 0.2,  target = 4 defectives
J_A ~ NegBin(4, 0.2)  =>  E[J_A] = 4/q = 4/0.2 = 20

Wald's equality: E[X_i] = q = 0.2
E[S_{J_A}] = 0.2 * 20 = 4  ✓
  (exactly 4 defectives found, by construction)
```

b) Good items inspected:

```
G = J_A - S_{J_A}  (total inspected minus defectives)
E[G] = E[J_A] - E[S_{J_A}] = 20 - 4 = 16 good items

Long-run fraction good = E[G]/E[J_A] = 16/20 = 0.8 = p  ✓
```

c) Stopping rule B — target 3 defectives, then 6:

```
q' = 1/4,  target k = 3
E[J_B] = k/q' = 3/(1/4) = 12

[Verify: Wald's equality  E[S_{J_B}] = (1/4)*12 = 3  ✓]

For target k = 6 (same q' = 1/4):
E[J_B] = 6/(1/4) = 24  (scales linearly with target)

General rule: E[J] = target / E[X] = k/q'
```